

KijaniKiosk

End-to-End Delivery Capstone

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Infrastructure-First



Problem & Scope

Problem

Deliver a repeatable, production-oriented KijaniKiosk platform with automated delivery, infrastructure, runtime controls and observability.

Track

Track A — Infrastructure-First

In scope

Infrastructure as Code • configuration management • CI/CD • Kubernetes runtime • monitoring • serverless payment flow

Out of scope

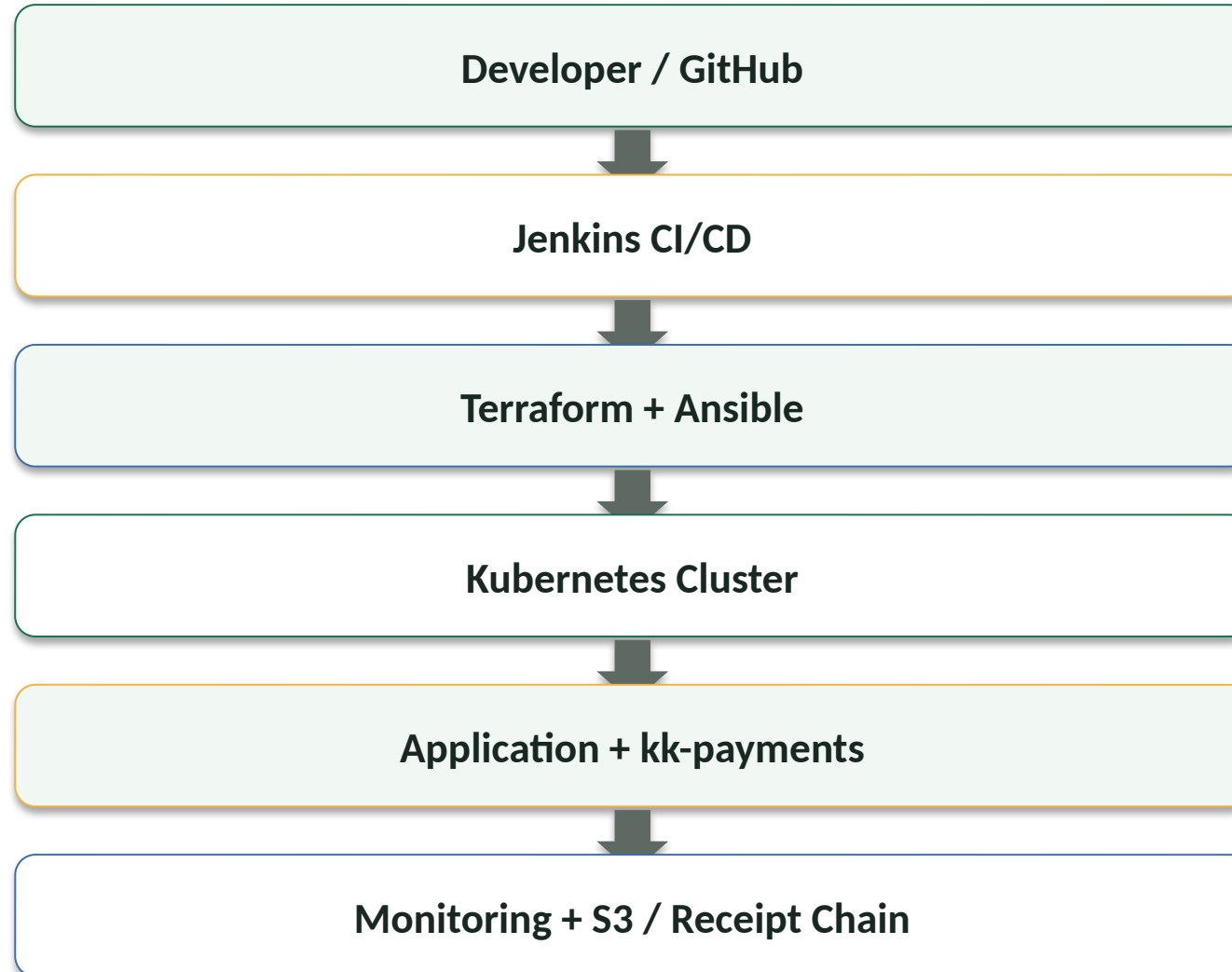
Major application feature development and full enterprise security hardening.

Success criteria

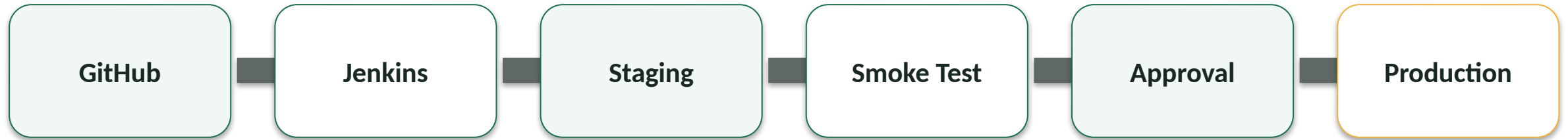
Reproducible infrastructure • reliable deployments • smoke-test validation • observable workloads • documented decisions and gaps

Architecture

Infrastructure-first delivery from source control to Kubernetes workloads and supporting services.



Delivery Pipeline



Automate

Build, test and package consistently.

Validate

Smoke tests catch deployment/runtime failures.

Control

Approval gates the production promotion.

Infrastructure & Runtime

Terraform

Provision infrastructure consistently and repeatably.

Ansible

Configure hosts and supporting runtime dependencies.

Kubernetes

Run application workloads with declarative manifests.

ConfigMaps

Separate environment configuration from application images.

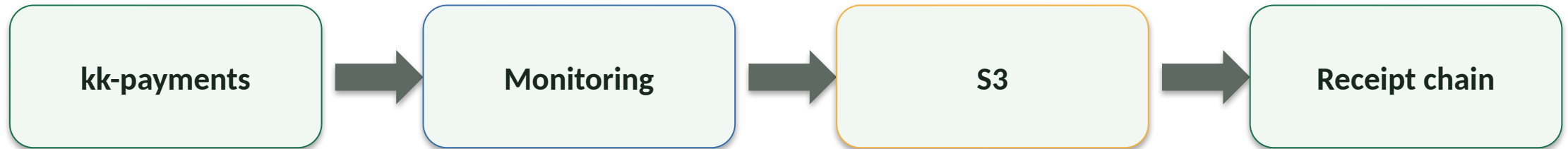
Probes

Readiness/liveness checks protect traffic and recovery.

Resources

CPU/memory requests and limits improve runtime control.

Monitoring & Serverless



Observe

Capture service health and operational signals.

Persist

Store receipt artifacts in S3.

Trace

Maintain a clear payment-to-receipt chain.

Key Technical Decision

Trade-off: simplicity and reuse vs environment-specific customization.

Option A — Single reusable Deployment manifest

Pros: less duplication, easier maintenance, consistent behavior.
Cons: environment differences require controlled configuration.

Option B — Separate staging/production manifests

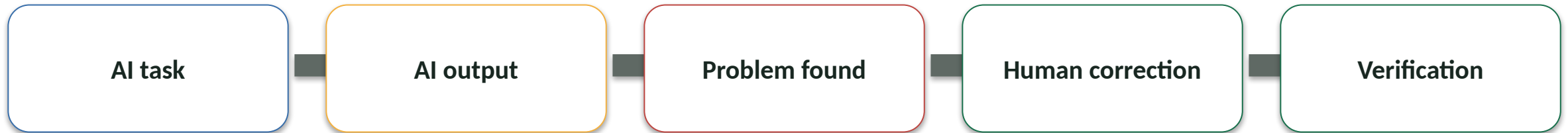
Pros: explicit environment-specific settings.
Cons: duplication, drift risk and more files to maintain.

Chosen: Single reusable Deployment manifest

Environment-specific values are externalized through configuration rather than duplicated manifests.

AI Governance

AI was treated as an assistant, not an authority.



Governance principle

Record what AI produced, identify errors, apply human correction, then verify the final result against the project requirements.

Peer Feedback

Three concrete issues → concrete improvements.

Issue 1

Documentation / presentation needed clearer structure. Improvement: standardized README and slide structure.

Issue 2

Deployment flow needed stronger validation. Improvement: explicit smoke-test and approval stages.

Issue 3

Production risks needed to be stated more specifically. Improvement: document each gap with risk and remediation.

Production Gaps

Specific gaps, risks and actionable remediation.

GAP

Single Kubernetes cluster

RISK

Cluster failure affects all environments

REMEDIATION

Separate production cluster with managed control plane

GAP

Limited production-grade secrets strategy

RISK

Sensitive configuration may be exposed or difficult to rotate

REMEDIATION

Adopt a managed secrets solution with rotation and least privilege

GAP

Limited disaster recovery

RISK

Recovery from infrastructure/data loss may be slow

REMEDIATION

Add tested backups, recovery procedures and restore drills